

Figure 4: Cylindrical robot simulated in Simbad

downside is that this project is no more being developed; the latest version, Simbad 1.4, was released in July 2007.

Player Project

Though Simbad will have good interfacing of sonars, lasers, etc, you are bound to find restrictions on geometrical shapes, and the lack of a modular structure to incorporate any type of robot.

Voilà! Enter Player Project, <http://playerstage.sourceforge.net/>. Elements of Player Project have been discussed in Part 1 and Part 2 of this series. Player Project started off as a union of three software: Player (the server for controlling robots), Stage (2.5D simulations) and Gazebo (3D simulations). Later, both Stage and Gazebo were developed as standalones, and also incorporated into Robot Operating System (ROS). The great advantage that Player Project allows for is that the code used in simulation can also be used in real robots to provide for similar behaviour. Thus, Player Project assimilates the two interfaces of control and simulation, and nearly all current-day robotics platforms attempt to do so.

Other mentions, away from the open source realm, are Microsoft Robotics Developers Studio, <http://www.microsoft.com/robotics/> and Webots, <http://www.cyberbotics.com/overview>. Both are wonderful software, though not in the spirit of open source.

The truth is out there!

With the development of Robot Operating System, there was a surge in such platforms. One hurdle of Player Project was the issues with installation and stability. ROS structured various such robot software into an OS-like organisation. At the time of writing, ROS Fuerte (version 1.8.X) had been released,

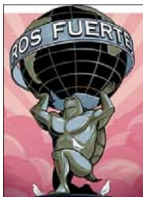


Figure 5: ROS Fuerte release



Figure 6: Darwin robot in Webot

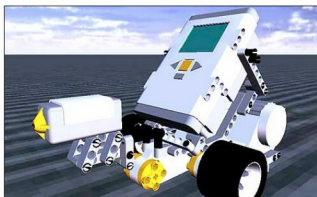


Figure 7: LEGO Mindstorms simulated in Microsoft Robotics Developers Studio

which allowed for a common platform for software such as Stage, Gazebo, Urbi, support for Microsoft Kinect, support for ASUS Xtion, bindings into MORSE, etc.

Other than ROS, enthusiasts can find their way to a host of such open source robotics software platforms. An incomplete list would be:

1. Player Project—<http://playerstage.sourceforge.net/>
2. MORSE—<http://www.openrobots.org/wiki/morse/>
3. Peekabot—<http://www.peekabot.org/>
4. YARP—<http://eris.liralab.it/yarp/>
5. MRPT—<http://www.mrpt.org/>
6. Carmen—<http://carmen.sourceforge.net/home.html>
7. Simbad—<http://simbad.sourceforge.net/>
8. Robocode—<http://robocode.sourceforge.net/>
9. Rossum's Playhouse—<http://rossum.sourceforge.net/sim.html>

sim.html 

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